CSC311 Project Idea V2

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Goal:

Given a partially observed 3D scan (occluded, truncated, or single-view), is it possible to reconstruct the missing geometry, and does a linear or nonlinear latent space recover it better? This is done by learning the compressed shape space using PCA and a small autoencoder (CAE), then we use a probability model over that space to fill in the missing geometry. Pretty much what modern 3D gen ai does built with classical ML algorithms.

Expanded:

If a self driving car's LIDAR only captures the front half of a parked car because the rest is occluded, can you reconstruct the full shape from that partial scan only?
The way we approach this problem is by learning a compressed representation of what objects in a particular category look like. Then we compare two ways of learning its representation, PCA for linear, and CAE for non-linear. ONce we have compressed the space, we fit a probability model over it using the EM algorithms (to iteratively solve its parameters)
Then, at test time, given only a partial scan, we search for the compressed code that best explains what we can see while still looking like a plausible object according to that probability model, and decode that back into a full 3D shape. We evaluate how close the reconstructed, hidden part is to the real thing, and we compare the linear approach against nonlinear one, including how much that probability model helps vs guessing blindly (or without one).
The bigger picture, or goal, is that this is a lightweight classical version of what modern 3D gen ai models do with more expensive methods. We want to find out if we can get away with cheaper methods from this course as opposed to large NN.

Formal Framing:

We address 3D shape completion from partial observations, relevant to AR scanning, robotic grasping, and occlusion-robust perception, by unifying linear (PCA) and nonlinear (autoencoder) shape representations under a shared completion objective that balances matching the visible geometry against staying plausible under the shapes the model has learned (a maximum a posteriori, or MAP, objective), with a Gaussian mixture prior fit via EM.

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We zero out every voxel the mask marks as unobserved, leaving only the visible portion of the shape as input, the observed partial input. Then, given a decoder (either PCA or AE), completion means finding the latent code that best explains what's visible while remaining plausible under the shapes the model has seen.

Simplified Work Flow:

Partial 3D input -> PCA completion (linear) and AE completion (non-linear) -> completed latent code -> decode full shape -> evaluate on masked region

Evaluation:

- ***Completion accuracy on the occluded region***: masked SDF error, PCA vs. AE.
- ***Prior ablation***: run completion with λ=0 (no GMM prior, pure data-fit) vs. λ>0, to directly test whether the prior helps
- ***Solve time***: closed-form vs. iterative
- ***Qualitative***: rendered completions next to ground truth, a handful of examples, and marching cubes.

Data set (already received approval):

https://huggingface.co/datasets/ShapeNet/ShapeNetCore
Will most likely make use of the chairs category, and possibly sofas. Our choice will depend on the loss of structural elements that may occur at practical grid resolutions during marching-cubes reconstruction.

Relevant Papers (for lit review):

AutoSDF (display of modern methods): https://arxiv.org/pdf/2203.09516
3DShape2VecSet: https://arxiv.org/pdf/2301.11445
Representing 3D Shapes with Probabilistic Directed Distance Fields: https://arxiv.org/pdf/2112.05300
3D ShapeNets: https://arxiv.org/pdf/1406.5670
A morphable model for the synthesis of 3D faces (fitting latent shape coefficients to partial/noisy observations under a Gaussian shape prior) https://dl.acm.org/doi/epdf/10.1145/311535.311556
Face Recognition Using Eigenfaces:
https://www.mit.edu/~9.54/fall14/Classes/class10/Turk%20Pentland%20Eigenfaces.pdf